

Task: 10

15 Marks

Lab Task 1

Create a class called Triangle that is derived from Shape. There are attributes called base (double), height (double) for this class. It inherits color from Shape. It has its own default constructor, and overloaded constructor. Both constructors call the base constructors. This class also contains the setter and getter functions for its attributes. Triangle class overrides the area() function and return the area of a triangle by calculating: $\frac{1}{2} (\text{base} * \text{height})$. Finally, it overrides the draw () function and displays all attributes and their values.

Create another class called Circle that is derived from Shape. There is attributes called radius (double) for this class. It inherits color from Shape. It has its own default constructor, and an overloaded constructor. Both constructors call the base constructors. This class also contains the setter and getter functions for its attributes. Circle class overrides the area () function and return the area of a circle by calculating: $\pi * \text{radius}^2$. It also contains a function called circumference () that returns the circumference ($2 * \pi * \text{radius}$) and another function called diameter () that returns the diameter ($2 * \text{radius}$) of a circle. Finally, it overrides the draw () function and displays all attributes and their values.

Write a main () program that creates an array of pointers to Shape. In a loop, ask the user for data about a particular triangle or circle, and use new to create an object of type Triangle or Circle to hold the data. Put the pointer to the object in the array. When the user has finished entering the data for all triangles and circles, display the resulting data for all the circles and triangles entered, using a for loop and a single statement such as:

pubarr[j] -> draw();

to display the data from each object in the array.

Lab Task 2

Design a Ship class that has the following members:

- A member variable for the name of the ship (a string)
- A member variable for the year that the ship was built (a string)
- A constructor and appropriate accessors and mutators
- A virtual print function that displays the ship's name and the year it was built.

Design a CruiseShip class that is derived from the Ship class. The CruiseShip class should have the following members:

- A member variable for the maximum number of passengers (an int)
- A constructor and appropriate accessors and mutators
- A print function that overrides the print function in the base class. The CruiseShip class's print function should display only the ship's name and the maximum number of passengers.

Design a CargoShip class that is derived from the Ship class. The CargoShip class should have the following members:

- A member variable for the cargo capacity in tonnage (an int)
- A constructor and appropriate accessors and mutators
- A print function that overrides the print function in the base class. The CargoShip class's print function should display only the ship's name and the ship's cargo capacity.

Demonstrate the classes in a program that has an array of Ship pointers. The array elements should be initialized with the addresses of dynamically allocated Ship, CruiseShip, and CargoShip objects. The program should then step through the array, calling each object's print function.

Lab Task 3

Create a class Person. This class contains three protected data members: name (string), age (int), and isMale (bool). Write the default and three argument overloaded constructors. Write setters and getters for all data members. Finally, write a pure virtual function called showData() with a return type of void.

Secondly, write a class called Employee that is publically inherited from Person. This class has two protected data members: employerName (string), and hourlyWage (double). Write setter and getter functions for all the data members. Also, write the default and overloaded constructors for this class. Then, write a function called showData() that displays all the data in an Employee object.

Finally, write a class called Teacher that is publically inherited from the Person. This class has three private data members: address (string), contactNumber (string), and teachesGrade (int). Write setter and getters for all the data members. Also write the default and overloaded constructors for this class that initializes all the data members of this class, including the inherited members. These constructors should also call the overloaded constructors of its parent.

Write the showData () function that displays all the data for this class and main () program and create an array of pointers to Person. Create at least two objects each of Employee and Teacher using various constructors. Assign the pointers of these objects to the array. Write a loop and display all the data stored in these objects by calling their showdata() function.

Lab Task 4

Write templates for the two functions minimum and maximum. The minimum function should accept two arguments and return the value of the argument that is the lesser of the two. The maximum function should accept two arguments and return the value of the argument that is the greater of the two. Design a simple driver program that demonstrates the templates with various data types.

Lab Task 5

Write a template function called swap () that interchanges the values of the two arguments sent to it. (You will probably want to pass these arguments by reference/pointers). Write a main() program to exercise the function with several types.

Lab Task 6

Write a function template that accepts an argument and returns its absolute value. The absolute value of a number is its value with no sign. For example, the absolute value of -5 is 5, and the absolute value of 2 is 2. Test the template in a simple driver program.